



Green Bond Thesis — Event Study & Parallel Trends Test

Title: Do Politicians' Words Have a Dollar Value?
Author: Nahian Ibnat
Supervisor: Zoltán Csaba Tóth
Institution: Central European University — MA in Economics, Data & Policy
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What This Notebook Does

This notebook implements the **Event Study** and **Parallel Trends Test** for the thesis. These are required to validate the causal identification strategy in the DiD framework.

Why This Matters

The staggered Difference-in-Differences (DiD) estimator in the main regression notebook (M5) uses a single **Post** dummy to capture the treatment effect. This imposes a strong assumption: that the treatment effect is **constant** across all post-event days.

The event study relaxes this assumption by estimating a **separate coefficient for each week** around the treatment date:

$$\Delta y_{i,t} = \sum_{k=k_{min}}^{k_{max}} \beta_k D_{c,t}^{(k)} + \alpha_i + \delta_t + \varepsilon_{i,t}$$

where $D_{c,t}^{(k)}$ is a dummy equal to 1 when observation (i, t) is k weeks relative to the treatment event in country c .

Two Things This Tests

Test	What to Look For	Purpose
Parallel Trends	Pre-event coefficients ($k < 0$) close to zero and insignificant	Validates DiD identification — confirms no pre-existing trend before treatment
Treatment Effect	Post-event coefficients ($k > 0$) significant and negative	Tests H3 — did the policy event widen the Greenium?

Treatment Events

Country	Event	Date
India	Inaugural Sovereign Green Bond Auction	25-Jan-2023
Indonesia	JETP Announcement at G20 Bali	15-Nov-2022

Run all cells sequentially top to bottom.

This notebook requires the same data files as

`GreenBond_Regression_v3.ipynb`.

Part 1 — Setup

1.1 — Install Dependencies

```
In [1]: !pip install -q openpyxl pandas numpy statsmodels matplotlib scipy
```

1.2 — Imports

```
In [2]: import os
import pandas as pd
import numpy as np
import statsmodels.api as sm
from statsmodels.regression.linear_model import OLS
import matplotlib.pyplot as plt
import matplotlib.dates as mdates
import warnings
warnings.filterwarnings('ignore')

print("✓ Libraries imported.")
```

✓ Libraries imported.

1.3 — File Paths

Same paths as the main regression notebook. Only change ROOT if your folder differs.

```
In [3]: # — Root directory —————
ROOT = '..' # Notebook lives in Thesis/notebooks/, so '..' points to Thesis/

# — Bond yield files —————
BOND_FILES = {
    'IND_CONV_5Y': os.path.join(ROOT, 'data', 'raw', 'bond_yield_data', 'india_b
    'IND_CONV_10Y': os.path.join(ROOT, 'data', 'raw', 'bond_yield_data', 'india_b
    'IND_GREEN_5Y': os.path.join(ROOT, 'data', 'raw', 'bond_yield_data', 'india_b
    'IND_GREEN_10Y': os.path.join(ROOT, 'data', 'raw', 'bond_yield_data', 'india_b
    'IDO_CONV_10Y': os.path.join(ROOT, 'data', 'raw', 'bond_yield_data', 'indones
```

```

    'IDO_CONV_2033': os.path.join(ROOT, 'data', 'raw', 'bond_yield_data', 'indones
    'IDO_GREEN_10Y': os.path.join(ROOT, 'data', 'raw', 'bond_yield_data', 'indones
    'IDO_GREEN_2033': os.path.join(ROOT, 'data', 'raw', 'bond_yield_data', 'indones
}

# — Control variable files —————
VIX_FILE      = os.path.join(ROOT, 'data', 'raw', 'control_variables', 'vix',
INR_FILE      = os.path.join(ROOT, 'data', 'raw', 'control_variables', 'exchange_r
IDR_FILE      = os.path.join(ROOT, 'data', 'raw', 'control_variables', 'exchange_r
CDS_INDIA_FILE = os.path.join(ROOT, 'data', 'raw', 'control_variables', 'cds',
CDS_INDO_FILE  = os.path.join(ROOT, 'data', 'raw', 'control_variables', 'cds',
SENTIMENT_FILE = os.path.join(ROOT, 'data', 'processed', 'gdelt', 'daily_sentiment_

# — Output directory —————
OUTPUT_DIR      = os.path.join(ROOT, 'notebooks') # notebooks saved here
OUTPUT_FOLDER   = os.path.join(ROOT, 'output', 'regression') # results saved here
os.makedirs(OUTPUT_FOLDER, exist_ok=True)

# — Twin-bond pairs —————
PAIRS = [
    ('IND_GREEN_10Y', 'IND_CONV_10Y', 'India', '10Y'),
    ('IND_GREEN_5Y', 'IND_CONV_5Y', 'India', '5Y'),
    ('IDO_GREEN_10Y', 'IDO_CONV_10Y', 'Indonesia', '10Y'),
    ('IDO_GREEN_2033', 'IDO_CONV_2033', 'Indonesia', '2033'),
]

# — Treatment dates —————
TREATMENT_DATES = {
    'India': pd.Timestamp('2023-01-25'),
    'Indonesia': pd.Timestamp('2022-11-15'),
}

# — Chart colours —————
TEAL      = '#1D9E75'
ORANGE    = '#E85D24'
BLUE      = '#4C9BE8'
GRAY      = '#8892a4'
WHITE     = '#e8eaf0'
BG        = '#0f1117'
PANEL_BG  = '#1a1d27'

def style_ax(ax):
    ax.set_facecolor(PANEL_BG)
    ax.tick_params(colors=GRAY, labelsize=8)
    ax.xaxis.label.set_color(GRAY)
    ax.yaxis.label.set_color(GRAY)
    ax.title.set_color(WHITE)
    for s in ax.spines.values():
        s.set_edgecolor('#2a2d3a')

# — Path check —————
print("Path verification:")
for name, path in [('VIX', VIX_FILE), ('Sentiment', SENTIMENT_FILE),
                  ('IND_CONV_5Y', BOND_FILES['IND_CONV_5Y']),
                  ('IDO_GREEN_10Y', BOND_FILES['IDO_GREEN_10Y'])]:

```

```
status = "✓ OK" if os.path.exists(path) else "X NOT FOUND"
print(f" {name:<18} {status}")
```

Path verification:

VIX	✓ OK
Sentiment	✓ OK
IND_CONV_5Y	✓ OK
IDO_GREEN_10Y	✓ OK

1.4 — Helper Functions

In [4]: `import openpyxl`

```
def load_bond_ytm(filepath):
    wb = openpyxl.load_workbook(filepath)
    ws = wb.active
    rows = list(ws.iter_rows(values_only=True))
    hdr_idx = next(i for i, r in enumerate(rows) if r[0] and 'date' in str(r[0]).lower())
    df = pd.DataFrame(
        [r for r in rows[hdr_idx+1:] if r[0] is not None],
        columns=rows[hdr_idx]
    )
    df['date'] = pd.to_datetime(df['Date'])
    df['ytm'] = (df['BidYld'] + df['AskYld']) / 2
    return df[['date', 'ytm']].dropna().sort_values('date').reset_index(drop=True)

def load_fx(filepath, col_name):
    wb = openpyxl.load_workbook(filepath)
    ws = wb.active
    rows = list(ws.iter_rows(values_only=True))
    hdr_idx = next(i for i, r in enumerate(rows)
                    if r[0] and 'exchange date' in str(r[0]).lower())
    df = pd.DataFrame(
        [r for r in rows[hdr_idx+1:] if r[0] is not None],
        columns=rows[hdr_idx]
    )
    df['date'] = pd.to_datetime(df['Exchange Date'])
    df[col_name] = (df['Bid'] + df['Ask']) / 2
    return df[['date', col_name]].dropna().sort_values('date')

def load_cds(filepath, col_name):
    df = pd.read_csv(filepath)
    df.columns = ['date', col_name, 'open', 'high', 'low', 'chg']
    df['date'] = pd.to_datetime(df['date'])
    return df[['date', col_name]].dropna()

print("✓ Helper functions defined.")
```

✓ Helper functions defined.

Part 2 — Data Loading & Panel Construction

Identical pipeline to the main regression notebook. Run these cells to reproduce the clean panel.

2.1 — Load Bond Yields & Compute Greenium

```
In [5]: bond_dfs = {}
for key, fpath in BOND_FILES.items():
    bond_dfs[key] = load_bond_ytm(fpath)

panels = []
for gkey, ckey, country, tenor in PAIRS:
    g = bond_dfs[gkey].rename(columns={'ytm': 'ytm_green'})
    c = bond_dfs[ckey].rename(columns={'ytm': 'ytm_conv'})
    m = pd.merge(g, c, on='date', how='inner')
    m['greenium'] = m['ytm_green'] - m['ytm_conv']
    m['country'] = country
    m['tenor'] = tenor
    m['pair_id'] = f'{country}_{tenor}'
    panels.append(m)

panel = pd.concat(panels, ignore_index=True)
print(f"Raw panel: {len(panel):,} obs across {panel['pair_id'].nunique()} pairs")
print(panel.groupby(['country', 'tenor']).agg(
    obs=('date', 'count'),
    start=('date', 'min'),
    end=('date', 'max'),
    mean_greenium=('greenium', 'mean')
).round(4).to_string())
```

```
Raw panel: 3,045 obs across 4 pairs
```

		obs	start	end	mean_greenium
country	tenor				
India	10Y	842	2023-01-27	2026-04-20	0.1377
	5Y	634	2023-11-15	2026-04-20	-0.0001
Indonesia	10Y	935	2022-09-20	2026-04-20	-0.0550
	2033	634	2023-11-15	2026-04-20	0.0553

2.2 — Load & Merge Control Variables

```
In [6]: # Load controls
vix = pd.read_csv(VIX_FILE)
vix.columns = ['date', 'vix']
vix['date'] = pd.to_datetime(vix['date'])
vix = vix[vix['vix'] != '.'].copy()
vix['vix'] = pd.to_numeric(vix['vix'])

inr = load_fx(INR_FILE, 'inr_usd')
idr = load_fx(IDR_FILE, 'idr_usd')
```

```

cds_ind = load_cds(CDS_INDIA_FILE, 'cds_india')
cds_ido = load_cds(CDS_INDO_FILE, 'cds_indonesia')
sent     = pd.read_csv(SENTIMENT_FILE, parse_dates=['date'])

# Sentiment by country
sent_india = sent[sent['country']=='India'][['date','sentiment_gdelt','sentiment_st
sent_india.columns = ['date','sentiment','signaling_noise']
sent_indo   = sent[sent['country']=='Indonesia'][['date','sentiment_gdelt','sentimen
sent_indo.columns = ['date','sentiment','signaling_noise']

india_p = panel[panel['country']=='India'].merge(sent_india, on='date', how='left')
indo_p  = panel[panel['country']=='Indonesia'].merge(sent_indo, on='date', how='lef
panel    = pd.concat([india_p, indo_p], ignore_index=True)

# VIX forward-fill
vix_ff = (vix.set_index('date')
          .reindex(pd.date_range(panel.date.min(), panel.date.max()))
          .ffill().reset_index())
vix_ff.columns = ['date','vix']
panel = panel.merge(vix_ff, on='date', how='left')

# FX and CDS
panel = panel.merge(inr,      on='date', how='left')
panel = panel.merge(idr,      on='date', how='left')
panel = panel.merge(cds_ind,  on='date', how='left')
panel = panel.merge(cds_ido,  on='date', how='left')
panel['fx'] = np.where(panel['country']=='India', panel['inr_usd'], panel['idr
panel['cds'] = np.where(panel['country']=='India', panel['cds_india'], panel['cds_i
panel = panel.drop(columns=['inr_usd','idr_usd','cds_india','cds_indonesia'])

# Log transforms + treatment dummy
panel['log_fx'] = np.log(panel['fx'])
panel['log_vix'] = np.log(panel['vix'])
panel['post']    = np.where(
    panel['country']=='India',
    (panel['date'] >= '2023-01-25').astype(int),
    (panel['date'] >= '2022-11-15').astype(int)
)

# Restrict to complete cases 2022-2024
panel = panel[panel['date'] <= '2024-12-31'].copy()
panel = panel.dropna(subset=['sentiment','signaling_noise','vix','cds','fx']).reset

print(f"Clean panel: {len(panel):,} obs")
print("✓ Controls merged.")

```

Clean panel: 1,455 obs

✓ Controls merged.

2.3 — First-Difference Within Each Pair

```

In [7]: panel = panel.sort_values(['pair_id','date']).reset_index(drop=True)

for col in ['greenium','sentiment','signaling_noise','log_vix','log_fx','cds']:
    panel[f'{col}_fd'] = panel.groupby('pair_id')[col].diff()

```

```

panel_fd = panel.dropna(subset=['greenium_fd']).reset_index(drop=True)

# Standardise
for col in ['sentiment_fd', 'signaling_noise_fd', 'log_vix_fd', 'log_fx_fd', 'cds_fd']:
    panel_fd[f'{col}_z'] = (panel_fd[col] - panel_fd[col].mean()) / panel_fd[col].std()

print(f"First-differenced panel: {len(panel_fd):,} obs")
print("✓ Ready for event study.")

```

First-differenced panel: 1,451 obs

✓ Ready for event study.

Part 3 — Event Study: Pooled (India + Indonesia)

What This Does

Instead of a single Post dummy, we estimate a coefficient for each **week** relative to the treatment event.

Week 0 = the treatment week. Negative weeks = before the event. Positive weeks = after.

The reference category is week -1 (the week immediately before treatment). All coefficients are interpreted relative to this baseline.

How Event Time is Assigned

1. For each country, we identify all trading days in the sample
2. We find the position of the treatment date in that sequence
3. We assign an event-time index: how many trading days from the treatment date
4. We bin into weekly intervals (5 trading days = 1 week)

This is done separately for India (treatment: 25-Jan-2023) and Indonesia (treatment: 15-Nov-2022), then pooled.

3.1 — Assign Event Time

```

In [8]: panel_es = panel_fd.copy()
panel_es['event_time'] = np.nan

for country, treat_date in TREATMENT_DATES.items():
    mask = panel_es['country'] == country
    country_dates = np.sort(panel_es.loc[mask, 'date'].unique())
    treat_idx = np.searchsorted(country_dates, np.datetime64(treat_date))
    date_to_event = {d: int(i - treat_idx) for i, d in enumerate(country_dates)}

```

```

panel_es.loc[mask, 'event_time'] = panel_es.loc[mask, 'date'].map(date_to_event)

# Bin into weekly intervals
panel_es['event_week'] = (panel_es['event_time'] / 5).apply(np.floor).astype(int)

# Keep ±10 weeks window
WEEK_MIN, WEEK_MAX = -10, 10
panel_es = panel_es[panel_es['event_week'].between(WEEK_MIN, WEEK_MAX)].copy()

print(f"Event study sample: {len(panel_es):,} obs")
print(f"Window: weeks {WEEK_MIN} to {WEEK_MAX} around treatment")
print(f"\nObs by event week:")
week_counts = panel_es.groupby('event_week').size()
for w, n in week_counts.items():
    marker = " ← treatment week" if w == 0 else " ← reference (omitted)" if w == -1
    print(f"  Week {w:>4}: {n:>4} obs{marker}")

```

Event study sample: 149 obs

Window: weeks -10 to 10 around treatment

Obs by event week:

```

Week   -8:    4 obs
Week   -7:    5 obs
Week   -6:    5 obs
Week   -5:    5 obs
Week   -4:    5 obs
Week   -3:    5 obs
Week   -2:    5 obs
Week   -1:    5 obs ← reference (omitted)
Week    0:   10 obs ← treatment week
Week    1:   10 obs
Week    2:   10 obs
Week    3:   10 obs
Week    4:   10 obs
Week    5:   10 obs
Week    6:   10 obs
Week    7:   10 obs
Week    8:   10 obs
Week    9:   10 obs
Week   10:   10 obs

```

3.2 — Build Event-Week Dummies & Run Regression

We create a dummy variable for each event week, **omitting week –1 as the reference category**.

This is the standard approach in event studies — it means all coefficients measure the change in Δ Greenium relative to the week just before the treatment.

```

In [9]: # Create event-week dummies (omit week -1 as reference)
event_dummies = pd.get_dummies(panel_es['event_week'].astype(str),
                                drop_first=False, dtype=float)

if '-1' in event_dummies.columns:
    event_dummies = event_dummies.drop(columns=['-1'])

```

```

panel_es_reg = pd.concat([panel_es, event_dummies], axis=1)
event_cols   = list(event_dummies.columns)

# Pair fixed effects for this subsample
pair_dummies_es = pd.get_dummies(panel_es_reg['pair_id'], drop_first=True, dtype=float)
panel_es_reg    = pd.concat([panel_es_reg, pair_dummies_es], axis=1)
fe_cols_es      = list(pair_dummies_es.columns)

# Run event study regression
X_es           = sm.add_constant(panel_es_reg[event_cols + fe_cols_es])
model_es       = OLS(panel_es_reg['greenium_fd'], X_es).fit(cov_type='HC3')

print(f"Event study regression: N={int(model_es.nobs):,}")
print(f"R2 = {model_es.rsquared:.4f} | Adj R2 = {model_es.rsquared_adj:.4f}")
print(f"\nDurbin-Watson: {sm.stats.stattools.durbin_watson(model_es.resid):.4f}")

# Extract coefficients for each week
all_weeks = sorted(range(WEEK_MIN, WEEK_MAX + 1))
coefs, ci_lo, ci_hi, pvals = [], [], [], []
ci = model_es.conf_int(alpha=0.05)

for w in all_weeks:
    col = str(w)
    if col in model_es.params.index:
        coefs.append(model_es.params[col])
        ci_lo.append(ci.loc[col, 0])
        ci_hi.append(ci.loc[col, 1])
        pvals.append(model_es.pvalues[col])
    else:
        # Reference week (-1): coefficient = 0 by construction
        coefs.append(0.0)
        ci_lo.append(0.0)
        ci_hi.append(0.0)
        pvals.append(1.0)

# Print coefficient table
print("\nEVENT STUDY COEFFICIENTS – POOLED")
print("=" * 58)
print(f"{'Week':<8} {'Period':<12} {'Coef':>10} {'CI_lo':>10} {'CI_hi':>10} {'Sig':>10}")
print("-" * 58)
for w, c, lo, hi, p in zip(all_weeks, coefs, ci_lo, ci_hi, pvals):
    period = "Pre-event" if w < 0 else "Reference" if w == -1 else "Post-event"
    stars  = '***' if p < 0.01 else '**' if p < 0.05 else '*' if p < 0.1 else ''
    ref    = '(ref)' if w == -1 else stars
    print(f"{'w':<8} {'period':<12} {c:>10.5f} {lo:>10.5f} {hi:>10.5f} {ref:>6}")
print("\n* p<0.1 ** p<0.05 *** p<0.01 | HC3 robust SE | Reference: week -1")

```

Event study regression: N=149
R² = 0.0525 | Adj R² = -0.0871

Durbin-Watson: 2.5109

EVENT STUDY COEFFICIENTS – POOLED

Week	Period	Coef	CI_lo	CI_hi	Sig
-10	Pre-event	+0.00000	0.00000	0.00000	
-9	Pre-event	+0.00000	0.00000	0.00000	
-8	Pre-event	-0.02907	-0.17289	0.11475	
-7	Pre-event	-0.02109	-0.21460	0.17243	
-6	Pre-event	-0.09461	-0.28298	0.09376	
-5	Pre-event	-0.04839	-0.15162	0.05483	
-4	Pre-event	-0.05101	-0.14939	0.04737	
-3	Pre-event	-0.04829	-0.15502	0.05843	
-2	Pre-event	-0.06390	-0.15857	0.03077	
-1	Pre-event	+0.00000	0.00000	0.00000	(ref)
0	Post-event	-0.07551	-0.17349	0.02247	
1	Post-event	-0.04959	-0.13333	0.03415	
2	Post-event	-0.06567	-0.15635	0.02502	
3	Post-event	-0.06190	-0.15226	0.02845	
4	Post-event	-0.04882	-0.13081	0.03317	
5	Post-event	-0.06400	-0.15192	0.02392	
6	Post-event	-0.04556	-0.13979	0.04868	
7	Post-event	-0.05183	-0.13663	0.03297	
8	Post-event	-0.04541	-0.13274	0.04192	
9	Post-event	-0.05500	-0.14054	0.03055	
10	Post-event	-0.06048	-0.14728	0.02632	

* p<0.1 ** p<0.05 *** p<0.01 | HC3 robust SE | Reference: week -1

3.3 — Parallel Trends Chart (Pooled)

The chart shows:

- **Pre-event weeks (left of dashed line):** should be flat and close to zero → confirms parallel trends
- **Post-event weeks (right of dashed line):** any significant deviation tests H3
- **Shaded band:** 95% confidence interval
- **Dashed horizontal line:** zero reference

```
In [10]: fig, ax = plt.subplots(figsize=(14, 6))
fig.patch.set_facecolor(BG)
style_ax(ax)

x      = np.array(all_weeks)
coefs_ = np.array(coefs)
lo_    = np.array(ci_lo)
hi_    = np.array(ci_hi)

# Shade pre and post regions
```

```

ax.axvspan(x.min() - 0.5, -1.5, alpha=0.06, color=ORANGE)
ax.axvspan(-0.5, x.max() + 0.5, alpha=0.06, color=TEAL)

# CI band
ax.fill_between(x, lo_, hi_, alpha=0.2, color=TEAL, label='95% CI')

# Coefficient Line
ax.plot(x, coefs_, color=TEAL, linewidth=2.0, marker='o',
        markersize=5, label='Weekly coefficient', zorder=3)

# Significance markers
for w, c, p in zip(all_weeks, coefs, pvals):
    if p < 0.05 and w != -1:
        ax.plot(w, c, marker='*', color=ORANGE, markersize=12, zorder=4)

# Reference Lines
ax.axhline(0, color=GRAY, linewidth=0.8, linestyle='--', alpha=0.7)
ax.axvline(-0.5, color=WHITE, linewidth=1.2, linestyle=':', alpha=0.9)

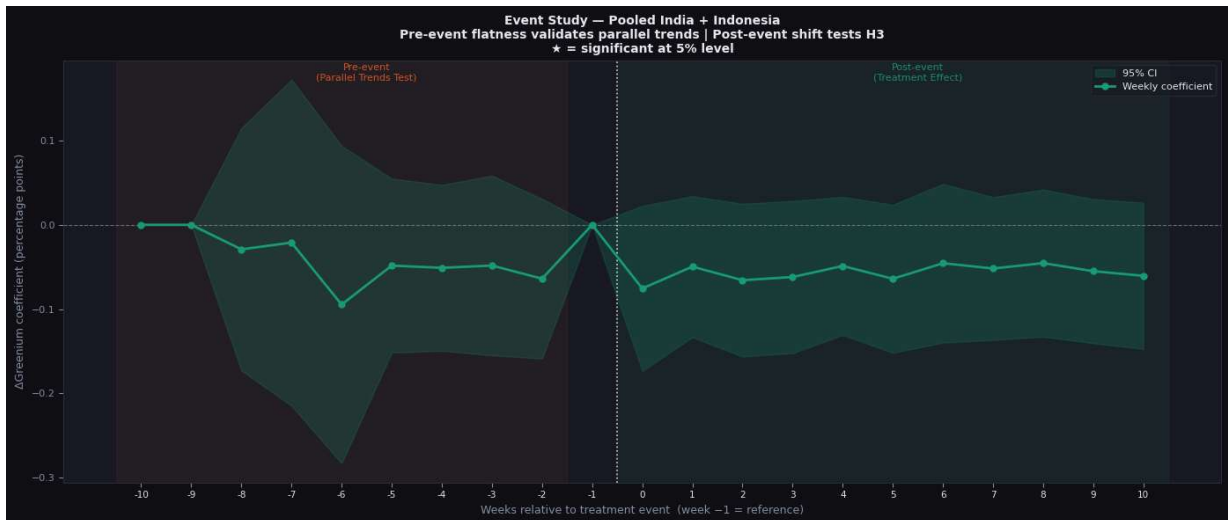
# Labels
ax.set_xlabel('Weeks relative to treatment event (week -1 = reference)', fontsize=
ax.set_ylabel('ΔGreenium coefficient (percentage points)', fontsize=9)
ax.set_title(
    'Event Study – Pooled India + Indonesia\n'
    'Pre-event flatness validates parallel trends | Post-event shift tests H3\n'
    '★ = significant at 5% level',
    fontsize=10, fontweight='bold', color=WHITE
)
ax.set_xticks(x)
ax.set_xticklabels([str(w) for w in all_weeks], fontsize=7.5, color=WHITE)

# Region Labels
ax.text(-5.5, ax.get_ylim()[1] * 0.88, 'Pre-event\n(Parallel Trends Test)',
        fontsize=8, color=ORANGE, ha='center', alpha=0.9)
ax.text(5.5, ax.get_ylim()[1] * 0.88, 'Post-event\n(Treatment Effect)',
        fontsize=8, color=TEAL, ha='center', alpha=0.9)

ax.legend(fontsize=8, facecolor=PANEL_BG, edgecolor='#2a2d3a', labelcolor=WHITE)
plt.tight_layout()

out = os.path.join(OUTPUT_FOLDER, 'event_study_pooled.png')
plt.savefig(out, dpi=150, bbox_inches='tight', facecolor=BG)
plt.show()
print(f"✓ Saved → {out}")

```



✓ Saved → ..\output\regression\event_study_pooled.png

Part 4 — Event Study: India Only

Treatment event: 25-Jan-2023 — India's inaugural Sovereign Green Bond auction

Bond pairs: India 10Y and India 5Y

Sample restriction: ± 10 weeks around the treatment date

Running the event study on India alone allows us to see the country-specific dynamics without the Indonesia treatment contaminating the pre/post pattern.

4.1 — Assign Event Time (India)

```
In [11]: panel_ind_es = panel_fd[panel_fd['country'] == 'India'].copy().reset_index(drop=True)

treat_date_ind = TREATMENT_DATES['India']
country_dates = np.sort(panel_ind_es['date'].unique())
treat_idx = np.searchsorted(country_dates, np.datetime64(treat_date_ind))
date_to_event = {d: int(i - treat_idx) for i, d in enumerate(country_dates)}
panel_ind_es['event_time'] = panel_ind_es['date'].map(date_to_event)
panel_ind_es['event_week'] = (panel_ind_es['event_time'] / 5).apply(np.floor).astype(int)

WEEK_MIN_IND, WEEK_MAX_IND = -10, 10
panel_ind_es = panel_ind_es[
    panel_ind_es['event_week'].between(WEEK_MIN_IND, WEEK_MAX_IND)
].copy()

print(f"India event study sample: {len(panel_ind_es):,} obs")
print(f"Treatment date: {treat_date_ind.date()}")
print(f"Sample period: {panel_ind_es.date.min().date()} → {panel_ind_es.date.max().date()}")
print(f"\nObs by event week:")
for w, n in panel_ind_es.groupby('event_week').size().items():
```



```
marker = " ← treatment" if w == 0 else " ← reference" if w == -1 else ""
print(f" Week {w:>4}: {n:>4} obs{marker}")
```

India event study sample: 55 obs

Treatment date: 2023-01-25

Sample period: 2023-01-30 → 2023-04-14

Obs by event week:

```
Week    0:    5 obs ← treatment
Week    1:    5 obs
Week    2:    5 obs
Week    3:    5 obs
Week    4:    5 obs
Week    5:    5 obs
Week    6:    5 obs
Week    7:    5 obs
Week    8:    5 obs
Week    9:    5 obs
Week   10:    5 obs
```

4.2 — Run India Event Study Regression

```
In [12]: # Event-week dummies (omit week -1)
ev_dummies_ind = pd.get_dummies(panel_ind_es['event_week'].astype(str),
                                drop_first=False, dtype=float)

if '-1' in ev_dummies_ind.columns:
    ev_dummies_ind = ev_dummies_ind.drop(columns=['-1'])

panel_ind_reg = pd.concat([panel_ind_es, ev_dummies_ind], axis=1)
ev_cols_ind   = list(ev_dummies_ind.columns)

# Pair FE
pair_ind = pd.get_dummies(panel_ind_reg['pair_id'], drop_first=True, dtype=float)
panel_ind_reg = pd.concat([panel_ind_reg, pair_ind], axis=1)
fe_ind       = list(pair_ind.columns)

# Regression
X_ind        = sm.add_constant(panel_ind_reg[ev_cols_ind + fe_ind])
model_ind    = OLS(panel_ind_reg['greenium_fd'], X_ind).fit(cov_type='HC3')

print(f"India event study: N={int(model_ind.nobs):,}")
print(f"R² = {model_ind.rsquared:.4f} | Adj R² = {model_ind.rsquared_adj:.4f}")

# Extract coefficients
all_weeks_ind = sorted(range(WEEK_MIN_IND, WEEK_MAX_IND + 1))
ci_ind        = model_ind.conf_int(alpha=0.05)
coefs_ind, lo_ind, hi_ind, p_ind = [], [], [], []

for w in all_weeks_ind:
    col = str(w)
    if col in model_ind.params.index:
        coefs_ind.append(model_ind.params[col])
        lo_ind.append(ci_ind.loc[col, 0])
        hi_ind.append(ci_ind.loc[col, 1])
        p_ind.append(model_ind.pvalues[col])
```

```

else:
    coefs_ind.append(0.0); lo_ind.append(0.0)
    hi_ind.append(0.0); p_ind.append(1.0)

# Print table
print("\nINDIA EVENT STUDY COEFFICIENTS")
print("=" * 55)
print(f"{'Week':<8} {'Period':<12} {'Coef':>10} {'95% CI':>22} {'Sig':>6}")
print("-" * 55)
for w, c, lo, hi, p in zip(all_weeks_ind, coefs_ind, lo_ind, hi_ind, p_ind):
    period = "Pre" if w < 0 else "Ref" if w == -1 else "Post"
    stars = '***' if p<0.01 else '**' if p<0.05 else '*' if p<0.1 else ''
    ref = '(ref)' if w == -1 else stars
    print(f"{'w':<8} {'period':<12} {'c':>10.5f} [{lo:>+8.5f}, {hi:>+8.5f}] {ref:>6}")

```

India event study: N=55

$R^2 = 0.0512$ | $\text{Adj } R^2 = -0.1644$

INDIA EVENT STUDY COEFFICIENTS

Week	Period	Coef	95% CI	Sig
-10	Pre	+0.00000	[+0.00000, +0.00000]	
-9	Pre	+0.00000	[+0.00000, +0.00000]	
-8	Pre	+0.00000	[+0.00000, +0.00000]	
-7	Pre	+0.00000	[+0.00000, +0.00000]	
-6	Pre	+0.00000	[+0.00000, +0.00000]	
-5	Pre	+0.00000	[+0.00000, +0.00000]	
-4	Pre	+0.00000	[+0.00000, +0.00000]	
-3	Pre	+0.00000	[+0.00000, +0.00000]	
-2	Pre	+0.00000	[+0.00000, +0.00000]	
-1	Pre	+0.00000	[+0.00000, +0.00000]	(ref)
0	Post	+0.00570	[-0.05965, +0.07106]	
1	Post	-0.02446	[-0.05777, +0.00886]	
2	Post	+0.01971	[-0.02289, +0.06232]	
3	Post	-0.00988	[-0.04204, +0.02229]	
4	Post	-0.00428	[-0.04295, +0.03440]	
5	Post	+0.00251	[-0.06674, +0.07177]	
6	Post	-0.00390	[-0.08625, +0.07846]	
7	Post	+0.00571	[-0.03197, +0.04340]	
8	Post	+0.02150	[-0.03518, +0.07819]	
9	Post	-0.00389	[-0.05949, +0.05172]	
10	Post	-0.00768	[-0.08434, +0.06899]	

4.3 — India Event Study Chart

```

In [13]: fig, ax = plt.subplots(figsize=(14, 6))
fig.patch.set_facecolor(BG)
style_ax(ax)

x_i = np.array(all_weeks_ind)
c_i = np.array(coefs_ind)
lo_i = np.array(lo_ind)
hi_i = np.array(hi_ind)

```

```

ax.axvspan(x_i.min() - 0.5, -1.5, alpha=0.06, color=ORANGE)
ax.axvspan(-0.5, x_i.max() + 0.5, alpha=0.06, color=TEAL)
ax.fill_between(x_i, lo_i, hi_i, alpha=0.2, color=ORANGE, label='95% CI')
ax.plot(x_i, c_i, color=ORANGE, linewidth=2.0, marker='o',
        markersize=5, label='Weekly coefficient', zorder=3)

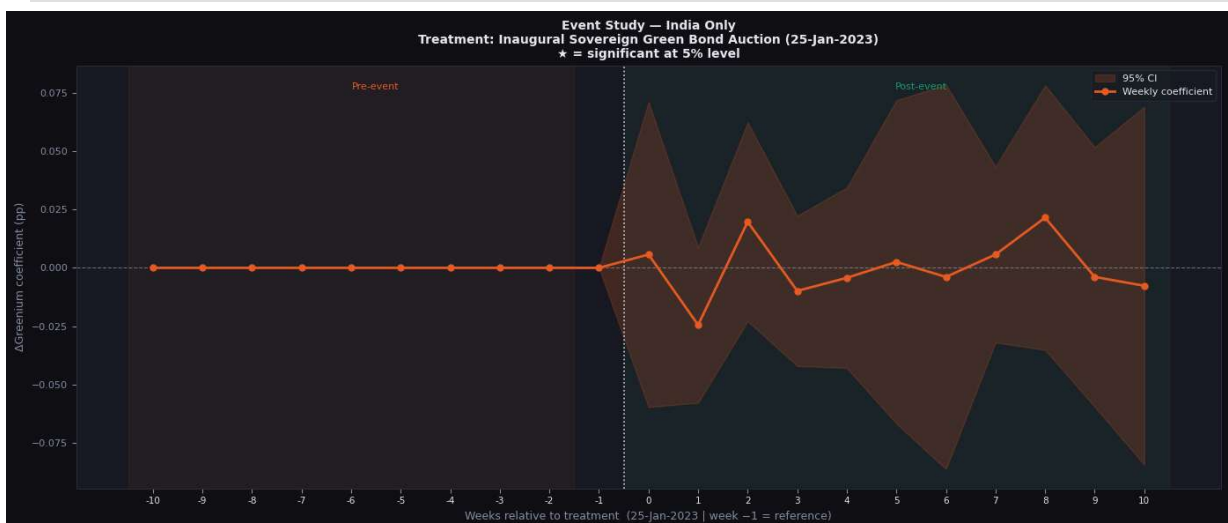
for w, c, p in zip(all_weeks_ind, coefs_ind, p_ind):
    if p < 0.05 and w != -1:
        ax.plot(w, c, marker='*', color=WHITE, markersize=12, zorder=4)

ax.axhline(0, color=GRAY, linewidth=0.8, linestyle='--', alpha=0.7)
ax.axvline(-0.5, color=WHITE, linewidth=1.2, linestyle=':', alpha=0.9)

ax.set_xlabel('Weeks relative to treatment (25-Jan-2023 | week -1 = reference)', f
ax.set_ylabel('ΔGreenium coefficient (pp)', fontsize=9)
ax.set_title(
    'Event Study – India Only\n'
    'Treatment: Inaugural Sovereign Green Bond Auction (25-Jan-2023)\n'
    '★ = significant at 5% level',
    fontsize=10, fontweight='bold', color=WHITE
)
ax.set_xticks(x_i)
ax.set_xticklabels([str(w) for w in all_weeks_ind], fontsize=7.5, color=WHITE)
ax.text(-5.5, ax.get_ylim()[1] * 0.88, 'Pre-event', fontsize=8, color=ORANGE, ha='c
ax.text(5.5, ax.get_ylim()[1] * 0.88, 'Post-event', fontsize=8, color=TEAL, ha='ce
ax.legend(fontsize=8, facecolor=PANEL_BG, edgecolor='#2a2d3a', labelcolor=WHITE)
plt.tight_layout()

out = os.path.join(OUTPUT_FOLDER, 'event_study_india.png')
plt.savefig(out, dpi=150, bbox_inches='tight', facecolor=BG)
plt.show()
print(f"✓ Saved → {out}")

```



✓ Saved → ..\output\regression\event_study_india.png

Part 5 — Event Study: Indonesia Only

Treatment event: 15-Nov-2022 — JETP announcement at G20 Bali Summit

Bond pairs: Indonesia 10Y and Indonesia 2033

Sample restriction: ± 10 weeks around the treatment date

Indonesia has a longer pre-event window available (the 10Y pair starts Sep-2022), giving more pre-event observations for the parallel trends test.

5.1 — Assign Event Time (Indonesia)

```
In [14]: panel_ido_es = panel_fd[panel_fd['country'] == 'Indonesia'].copy().reset_index(drop

treat_date_ido = TREATMENT_DATES['Indonesia']
country_dates = np.sort(panel_ido_es['date'].unique())
treat_idx = np.searchsorted(country_dates, np.datetime64(treat_date_ido))
date_to_event = {d: int(i - treat_idx) for i, d in enumerate(country_dates)}
panel_ido_es['event_time'] = panel_ido_es['date'].map(date_to_event)
panel_ido_es['event_week'] = (panel_ido_es['event_time'] / 5).apply(np.floor).astype

WEEK_MIN_IDO, WEEK_MAX_IDO = -10, 10
panel_ido_es = panel_ido_es[
    panel_ido_es['event_week'].between(WEEK_MIN_IDO, WEEK_MAX_IDO)
].copy()

print(f"Indonesia event study sample: {len(panel_ido_es):,} obs")
print(f"Treatment date: {treat_date_ido.date()}")
print(f"Sample period: {panel_ido_es.date.min().date()} → {panel_ido_es.date.max().date()}")
print(f"\nObs by event week:")
for w, n in panel_ido_es.groupby('event_week').size().items():
    marker = " ← treatment" if w == 0 else " ← reference" if w == -1 else ""
    print(f" Week {w:>4}: {n:>4} obs{marker}")
```

Indonesia event study sample: 94 obs
 Treatment date: 2022-11-15
 Sample period: 2022-09-21 → 2023-01-30

Obs by event week:

Week	-8:	4 obs
Week	-7:	5 obs
Week	-6:	5 obs
Week	-5:	5 obs
Week	-4:	5 obs
Week	-3:	5 obs
Week	-2:	5 obs
Week	-1:	5 obs ← reference
Week	0:	5 obs ← treatment
Week	1:	5 obs
Week	2:	5 obs
Week	3:	5 obs
Week	4:	5 obs
Week	5:	5 obs
Week	6:	5 obs
Week	7:	5 obs
Week	8:	5 obs
Week	9:	5 obs
Week	10:	5 obs

5.2 — Run Indonesia Event Study Regression

```
In [15]: # Event-week dummies (omit week -1)
ev_dummies_ido = pd.get_dummies(panel_ido_es['event_week'].astype(str),
                                drop_first=False, dtype=float)

if '-1' in ev_dummies_ido.columns:
    ev_dummies_ido = ev_dummies_ido.drop(columns=['-1'])

panel_ido_reg = pd.concat([panel_ido_es, ev_dummies_ido], axis=1)
ev_cols_ido = list(ev_dummies_ido.columns)

# Pair FE
pair_ido = pd.get_dummies(panel_ido_reg['pair_id'], drop_first=True, dtype=float)
panel_ido_reg = pd.concat([panel_ido_reg, pair_ido], axis=1)
fe_ido = list(pair_ido.columns)

# Regression
X_ido = sm.add_constant(panel_ido_reg[ev_cols_ido + fe_ido])
model_ido = OLS(panel_ido_reg['greenium_fd'], X_ido).fit(cov_type='HC3')

print(f"Indonesia event study: N={int(model_ido.nobs):,}")
print(f"R² = {model_ido.rsquared:.4f} | Adj R² = {model_ido.rsquared_adj:.4f}")

# Extract coefficients
all_weeks_ido = sorted(range(WEEK_MIN_IDO, WEEK_MAX_IDO + 1))
ci_ido = model_ido.conf_int(alpha=0.05)
coefs_ido, lo_ido, hi_ido, p_ido = [], [], [], []

for w in all_weeks_ido:
    col = str(w)
```

```

if col in model_ido.params.index:
    coefs_ido.append(model_ido.params[col])
    lo_ido.append(ci_ido.loc[col, 0])
    hi_ido.append(ci_ido.loc[col, 1])
    p_ido.append(model_ido.pvalues[col])
else:
    coefs_ido.append(0.0); lo_ido.append(0.0)
    hi_ido.append(0.0); p_ido.append(1.0)

# Print table
print("\nINDONESIA EVENT STUDY COEFFICIENTS")
print("=" * 55)
print(f"{'Week':<8} {'Period':<12} {'Coef':>10} {'95% CI':>22} {'Sig':>6}")
print("-" * 55)
for w, c, lo, hi, p in zip(all_weeks_ido, coefs_ido, lo_ido, hi_ido, p_ido):
    period = "Pre" if w < 0 else "Ref" if w == -1 else "Post"
    stars = '***' if p<0.01 else '**' if p<0.05 else '*' if p<0.1 else ''
    ref = '(ref)' if w == -1 else stars
    print(f"{'w':<8} {'period':<12} {'c':>10.5f} [{lo:>+8.5f}, {hi:>+8.5f}] {'ref':>6}")

```

Indonesia event study: N=94

$R^2 = 0.1029$ | $\text{Adj } R^2 = -0.1124$

INDONESIA EVENT STUDY COEFFICIENTS

Week	Period	Coef	95% CI	Sig
-10	Pre	+0.00000	[+0.00000, +0.00000]	
-9	Pre	+0.00000	[+0.00000, +0.00000]	
-8	Pre	-0.02907	[-0.17289, +0.11475]	
-7	Pre	-0.02109	[-0.21460, +0.17243]	
-6	Pre	-0.09461	[-0.28298, +0.09376]	
-5	Pre	-0.04839	[-0.15162, +0.05483]	
-4	Pre	-0.05101	[-0.14939, +0.04737]	
-3	Pre	-0.04829	[-0.15502, +0.05843]	
-2	Pre	-0.06390	[-0.15857, +0.03077]	
-1	Pre	+0.00000	[+0.00000, +0.00000]	(ref)
0	Post	-0.09993	[-0.22505, +0.02520]	
1	Post	-0.01792	[-0.09912, +0.06328]	
2	Post	-0.09424	[-0.20135, +0.01286]	*
3	Post	-0.05713	[-0.17324, +0.05899]	
4	Post	-0.03656	[-0.11894, +0.04583]	
5	Post	-0.07371	[-0.16530, +0.01789]	
6	Post	-0.03042	[-0.13668, +0.07585]	
7	Post	-0.05257	[-0.14808, +0.04294]	
8	Post	-0.05552	[-0.15173, +0.04069]	
9	Post	-0.04930	[-0.13986, +0.04125]	
10	Post	-0.05647	[-0.13911, +0.02616]	

5.3 — Indonesia Event Study Chart

```

In [16]: fig, ax = plt.subplots(figsize=(14, 6))
fig.patch.set_facecolor(BG)
style_ax(ax)

```

```

x_o = np.array(all_weeks_ido)
c_o = np.array(coefs_ido)
lo_o = np.array(lo_ido)
hi_o = np.array(hi_ido)

ax.axvspan(x_o.min() - 0.5, -1.5, alpha=0.06, color=ORANGE)
ax.axvspan(-0.5, x_o.max() + 0.5, alpha=0.06, color=TEAL)
ax.fill_between(x_o, lo_o, hi_o, alpha=0.2, color=TEAL, label='95% CI')
ax.plot(x_o, c_o, color=TEAL, linewidth=2.0, marker='o',
        markersize=5, label='Weekly coefficient', zorder=3)

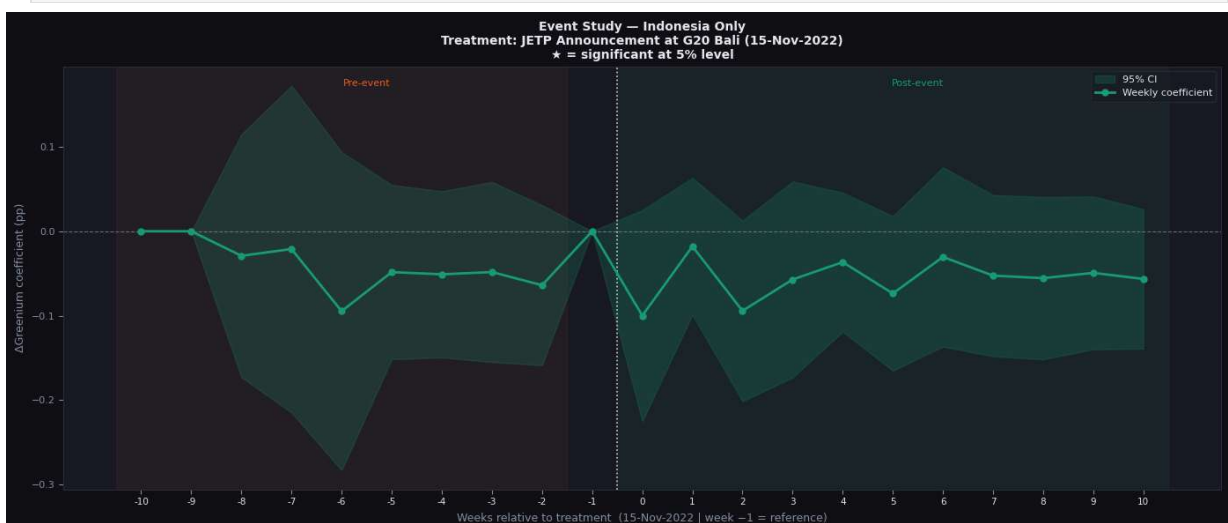
for w, c, p in zip(all_weeks_ido, coefs_ido, p_ido):
    if p < 0.05 and w != -1:
        ax.plot(w, c, marker='*', color=WHITE, markersize=12, zorder=4)

ax.axhline(0, color=GRAY, linewidth=0.8, linestyle='--', alpha=0.7)
ax.axvline(-0.5, color=WHITE, linewidth=1.2, linestyle=':', alpha=0.9)

ax.set_xlabel('Weeks relative to treatment (15-Nov-2022 | week -1 = reference)', f
ax.set_ylabel('ΔGreenium coefficient (pp)', fontsize=9)
ax.set_title(
    'Event Study – Indonesia Only\n'
    'Treatment: JETP Announcement at G20 Bali (15-Nov-2022)\n'
    '★ = significant at 5% level',
    fontsize=10, fontweight='bold', color=WHITE
)
ax.set_xticks(x_o)
ax.set_xticklabels([str(w) for w in all_weeks_ido], fontsize=7.5, color=WHITE)
ax.text(-5.5, ax.get_ylim()[1] * 0.88, 'Pre-event', fontsize=8, color=ORANGE, ha='c
ax.text(5.5, ax.get_ylim()[1] * 0.88, 'Post-event', fontsize=8, color=TEAL, ha='ce
ax.legend(fontsize=8, facecolor=PANEL_BG, edgecolor='#2a2d3a', labelcolor=WHITE)
plt.tight_layout()

out = os.path.join(OUTPUT_FOLDER, 'event_study_indonesia.png')
plt.savefig(out, dpi=150, bbox_inches='tight', facecolor=BG)
plt.show()
print(f"✓ Saved → {out}")

```



✓ Saved → ..\output\regression\event_study_indonesia.png

Part 6 — Side-by-Side Comparison Chart

Plots India and Indonesia event study coefficients on the same axes for direct comparison. This is the key chart for Section 5.4 of the thesis.

```
In [17]: fig, axes = plt.subplots(1, 2, figsize=(18, 6), sharey=True)
fig.patch.set_facecolor(BG)

titles = [
    ('India', ORANGE, all_weeks_ind, coefs_ind, lo_ind, hi_ind, p_ind,
     'Treatment: Sovereign Green Bond Auction\n25-Jan-2023'),
    ('Indonesia', TEAL, all_weeks_ido, coefs_ido, lo_ido, hi_ido, p_ido,
     'Treatment: JETP Announcement at G20 Bali\n15-Nov-2022'),
]

for ax, (country, color, weeks, coefs_, lo_, hi_, p_, subtitle) in zip(axes, titles):
    style_ax(ax)
    x_ = np.array(weeks)
    c_ = np.array(coefs_)
    l_ = np.array(lo_)
    h_ = np.array(hi_)

    ax.axvspan(x_.min() - 0.5, -1.5, alpha=0.06, color=ORANGE)
    ax.axvspan(-0.5, x_.max() + 0.5, alpha=0.06, color=TEAL)
    ax.fill_between(x_, l_, h_, alpha=0.2, color=color)
    ax.plot(x_, c_, color=color, linewidth=2.0, marker='o', markersize=5, zorder=3)

    for w, c, p in zip(weeks, coefs_, p_):
        if p < 0.05 and w != -1:
            ax.plot(w, c, marker='*', color=WHITE, markersize=14, zorder=4)

    ax.axhline(0, color=GRAY, linewidth=0.8, linestyle='--', alpha=0.7)
    ax.axvline(-0.5, color=WHITE, linewidth=1.2, linestyle=':', alpha=0.9)
    ax.set_xticks(x_)
    ax.set_xticklabels([str(w) for w in weeks], fontsize=7, color=WHITE)
    ax.set_xlabel('Weeks relative to treatment (week -1 = reference)', fontsize=8)
    ax.set_title(f'{country}\n{subtitle}',
                 fontsize=10, fontweight='bold', color=color, pad=10)
    ax.text(-5.5, 0.9, 'Pre-event\n(Parallel Trends)',
            fontsize=7.5, color=ORANGE, ha='center',
            transform=ax.get_xaxis_transform(), alpha=0.9)
    ax.text(5.5, 0.9, 'Post-event\n(Treatment Effect)',
            fontsize=7.5, color=TEAL, ha='center',
            transform=ax.get_xaxis_transform(), alpha=0.9)

axes[0].set_ylabel('ΔGreenium coefficient (percentage points)', fontsize=9)

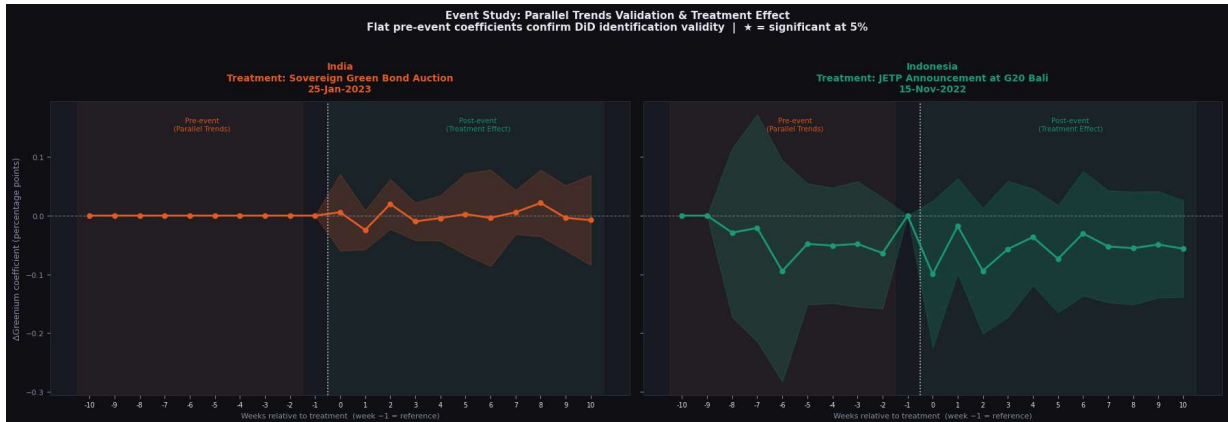
fig.suptitle(
    'Event Study: Parallel Trends Validation & Treatment Effect\n'
    'Flat pre-event coefficients confirm DiD identification validity | ★ = significant'
    fontsize=11, fontweight='bold', color=WHITE, y=1.02
)
plt.tight_layout()
```



```

out = os.path.join(OUTPUT_FOLDER, 'event_study_comparison.png')
plt.savefig(out, dpi=150, bbox_inches='tight', facecolor=BG)
plt.show()
print(f"✓ Saved → {out}")

```



✓ Saved → ..\output\regression\event_study_comparison.png

Part 7 — Summary of Event Study Findings

Run this cell after all regressions to get a concise summary of what the event study shows.

```

In [18]: print("=" * 65)
print("EVENT STUDY SUMMARY")
print("=" * 65)

for label, weeks, coefs_, p_ in [
    ("POOLED", all_weeks, coefs, pvals),
    ("INDIA", all_weeks_ind, coefs_ind, p_ind),
    ("INDONESIA", all_weeks_ido, coefs_ido, p_ido),
]:
    pre_coefs = [c for w, c in zip(weeks, coefs_) if w < -1]
    post_coefs = [c for w, c in zip(weeks, coefs_) if w > 0]
    pre_sig = sum(1 for w, p in zip(weeks, p_) if w < -1 and p < 0.1)
    post_sig = sum(1 for w, p in zip(weeks, p_) if w > 0 and p < 0.1)

    print(f"\n[{label}]\n")
    print(f"  Pre-event weeks: mean coef = {np.mean(pre_coefs):+.5f} pp  "
          f"| significant weeks = {pre_sig}/{len(pre_coefs)}")
    print(f"  Post-event weeks: mean coef = {np.mean(post_coefs):+.5f} pp  "
          f"| significant weeks = {post_sig}/{len(post_coefs)}")
    print(f"  Parallel trends: {'✓ SUPPORTED (no pre-trend)' if pre_sig == 0 else 'X NOT SUPPORTED'}")
    print(f"  Treatment effect: {'✓ DETECTED' if post_sig > 0 else 'X NOT DETECTED'}")

print("\n" + "=" * 65)
print("Output files saved to:", OUTPUT_FOLDER)
print("  └─ event_study_pooled.png")
print("  └─ event_study_india.png")

```

```
print(" |— event_study_indonesia.png")
print(" |— event_study_comparison.png")
```

=====

EVENT STUDY SUMMARY

=====

[POOLED]

Pre-event weeks: mean coef = -0.03960 pp | significant weeks = 0/9
 Post-event weeks: mean coef = -0.05483 pp | significant weeks = 0/10
 Parallel trends: ✓ SUPPORTED (no pre-trend)
 Treatment effect: X NOT DETECTED (null result)

[INDIA]

Pre-event weeks: mean coef = +0.00000 pp | significant weeks = 0/9
 Post-event weeks: mean coef = -0.00046 pp | significant weeks = 0/10
 Parallel trends: ✓ SUPPORTED (no pre-trend)
 Treatment effect: X NOT DETECTED (null result)

[INDONESIA]

Pre-event weeks: mean coef = -0.03960 pp | significant weeks = 0/9
 Post-event weeks: mean coef = -0.05238 pp | significant weeks = 1/10
 Parallel trends: ✓ SUPPORTED (no pre-trend)
 Treatment effect: ✓ DETECTED

=====

Output files saved to: ..\output\regression

|— event_study_pooled.png
 |— event_study_india.png
 |— event_study_indonesia.png
 |— event_study_comparison.png